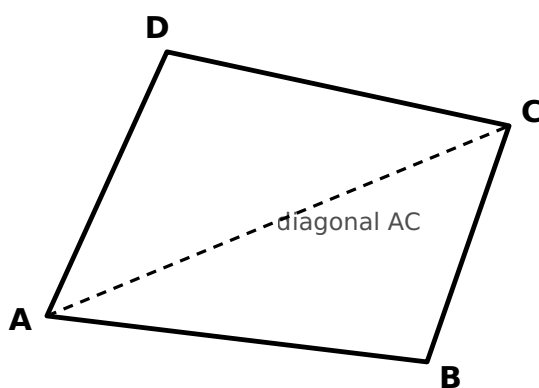


Angles in Quadrilaterals

Every quadrilateral's angles add up to 360° . Learn why, how to find a missing angle, and the polygon angle facts (interior, exterior, and number of diagonals) that the exemplar tests.

The angle sum is 360°

The rule: The four interior angles of **any** quadrilateral add up to 360° .

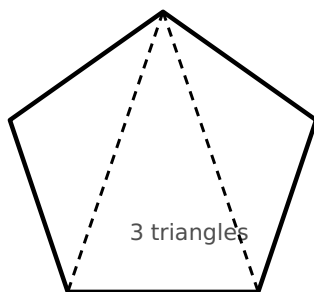


A diagonal cuts the quadrilateral into two triangles.

Why 360° ? Draw one diagonal. It splits the quadrilateral into **two triangles**. Each triangle's angles add to 180° , so together: $180^\circ + 180^\circ = 360^\circ$.

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

Polygon angle facts (used by the exemplar)



A polygon with n sides splits into $(n - 2)$ triangles. A pentagon $\rightarrow 3$.

- **Sum of interior angles** of an n -sided polygon = $(n - 2) \times 180^\circ$. (Quadrilateral: $(4 - 2) \times 180 = 360^\circ$.)

- **Sum of exterior angles** of any polygon = 360° (one at each vertex).
- **Regular polygon:** each exterior angle = $360^\circ/n$, and each interior angle = $180^\circ - (360^\circ/n)$.
- **Number of diagonals** of an n -sided polygon = $n(n-3)/2$. (Quadrilateral: $4 \times 1/2 = 2$.)

Prerequisite refresher: "Interior angle" is the angle inside a corner. An "exterior angle" is the turn you make at a corner, and an interior + its exterior = 180° .

Where marks slip away

Trap 1 — Using 180° for a quadrilateral. 180° is for a triangle. A quadrilateral is 360° .

Trap 2 — Ratio problems. If angles are in ratio 1:2:3:4, the parts add to 10, and 10 parts = 360° , so one part = 36° . Don't forget to multiply each ratio number by 36.

Trap 3 — Exterior-angle formula. For a regular polygon, number of sides = $360^\circ \div$ (one exterior angle). Divide into 360, don't multiply.

Trap 4 — Diagonals formula. It is $n(n-3)/2$, not $n(n-1)/2$. A quadrilateral has 2 diagonals, not 6.

Slip 5 — Units. Always write the degree symbol ($^\circ$) on angle answers.

Model answers — write them like this

Example A. In quadrilateral ABCD, $\angle A = 100^\circ$, $\angle B = 80^\circ$, $\angle C = 70^\circ$. Find $\angle D$.

Step 1: All four angles add to 360° : $\angle A + \angle B + \angle C + \angle D = 360^\circ$.

Step 2: $100^\circ + 80^\circ + 70^\circ + \angle D = 360^\circ$, so $250^\circ + \angle D = 360^\circ$.

Step 3: $\angle D = 360^\circ - 250^\circ = 110^\circ$.

$$\therefore \angle D = 110^\circ$$

Example B. The angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4. Find each angle.

Step 1: Let the angles be $1x$, $2x$, $3x$, $4x$. Their sum is 360° .

Step 2: $1x + 2x + 3x + 4x = 10x = 360^\circ$, so $x = 36^\circ$.

Step 3: Angles = 36° , 72° , 108° , 144° .

$$\therefore 36^\circ, 72^\circ, 108^\circ, 144^\circ$$

Example C. Each exterior angle of a regular polygon is 45° . How many sides has it?

Step 1: Sum of exterior angles = 360° , and they are all equal.

Step 2: Number of sides = $360^\circ \div 45^\circ = 8$.

$\therefore$ 8 sides (an octagon)

Example D. How many diagonals does a hexagon (6 sides) have?

Step 1: Use the formula $n(n - 3)/2$ with $n = 6$.

Step 2: $6 \times (6 - 3)/2 = 6 \times 3/2 = 9$.

$\therefore$ 9 diagonals

Next: [try the worksheet](#), then [check the answer key](#).